

David Roe’s Candidate Presentation for the Absolute Galois Group of $\mathbb{Q}_2$

A verification by the finite-quotient method

July 24, 2026

Abstract

We verify the candidate presentation for $G_{\mathbb{Q}_2}$ listed as David Roe’s candidate on the presentation website [3]. The verification is relative to the general local-cohomological and finite-target results of Roe–Turturean [1]: we prove that replacing their candidate word by Roe’s word preserves every source-side input to their boundary-framed finite-quotient induction. The main new point is the integral marking of the maximal pro-2 quotient. Its canonical Demushkin orientation is computed directly from Labute’s crossed-derivation characterization. The evaluated Fox matrix is then the matrix of [1, §5] with the two wild columns interchanged, and the Fox–Heisenberg pairing and determinant quadratic form are unchanged after the same interchange. The induction of [1, §§7–10] consequently gives equality of epimorphism counts to every finite group and hence the asserted profinite isomorphism.

Logical status. All calculations depending on Roe’s relator are proved below. The local Tate-duality, Gauss-sign, Fourier-inversion, target-induction, and profinite-reconstruction theorems are invoked from [1]; they are not reproved here.

Contents

1	The candidate and the replacement theorem	2
2	Tame quotient and unramified marking	2
3	The maximal pro-2 quotient	3
3.1	The relator and its cup form	3
3.2	The canonical orientation	4
3.3	Matching the full unramified character	5
4	The evaluated Fox complex	6
5	Fox–Heisenberg duality	8
5.1	The Stokes endpoint	8
5.2	The mixed Hessian	8
6	Quadratic determinant obstructions	9
7	Passage to every finite target	10
8	Audit summary and remaining formalization work	11

1 The candidate and the replacement theorem

We use the conventions

$$g^h = h^{-1}gh, \quad [g, h] = g^{-1}h^{-1}gh.$$

Let

$$\omega_2 = (1 \text{ in } \mathbf{Z}_2, 0 \text{ in } \mathbf{Z}_\ell \text{ for every odd prime } \ell) \in \widehat{\mathbf{Z}}.$$

Thus $\omega_2^2 = \omega_2$, and exponentiation by ω_2 extracts the 2-primary component of the procyclic group generated by an element. Put

$$\sigma_2 = \sigma^{\omega_2}, \quad a = (x_0^{-3}\tau)^{\omega_2}, \quad y_1 = x_1^{\sigma_2}, \quad c = [x_1, y_1]. \quad (1.1)$$

The two relators are

$$r_t = (\tau^\sigma)^{-1}\tau^2, \quad r_R = (x_0^\sigma)^{-1}a x_1^2 c. \quad (1.2)$$

Definition 1.1. Let F be the free profinite group on σ, τ, x_0, x_1 . A finite quotient of F is *R-admissible* if the distinguished tuple generates the quotient, the two words in [equation \(1.2\)](#) vanish, and the normal closure of the images of x_0, x_1 is a 2-group. Let Γ_R be the inverse limit of all *R-admissible* finite quotients. Equivalently, Γ_R is the profinite group with the displayed generators and relators, subject to the condition that $\langle\langle x_0, x_1 \rangle\rangle$ be pro-2.

This inverse-limit formulation is the same finite-quotient semantics used in [1, §2]; in particular it makes the “normal closure is pro-2” condition part of the presentation.

Theorem 1.2 (Replacement theorem). *Assume the local and finite-target results of [1]. Then the proof of [1, Theorem 1.2] remains valid with its candidate source replaced by Γ_R . Consequently*

$$\boxed{\Gamma_R \cong G_{\mathbf{Q}_2}}.$$

The proof occupies the rest of the note. We verify, in order:

1. the common tame quotient and full unramified character;
2. the fully marked maximal pro-2 quotient;
3. candidate deformation duality for every elementary characteristic-2 coefficient module;
4. equality of the candidate and local quadratic Gauss signs.

These are precisely the inputs isolated in [1, Corollary 6.19]. The target-side recursion then depends on no further feature of the source word; see [1, Theorem 8.17].

2 Tame quotient and unramified marking

Let $W_R = \overline{\langle\langle x_0, x_1 \rangle\rangle}$ in Γ_R .

Lemma 2.1. *The quotient by W_R is the standard tame group*

$$\mathcal{T} = \langle \sigma, \tau \mid \tau^\sigma = \tau^2 \rangle.$$

Moreover,

$$\nu_R(\sigma) = 1, \quad \nu_R(\tau) = \nu_R(x_0) = \nu_R(x_1) = 0$$

defines a continuous epimorphism $\nu_R : \Gamma_R \twoheadrightarrow \mathbf{Z}_2$, and its restriction to $\mathcal{T}$ is the standard geometric unramified character.

Proof. In a finite quotient of $\langle \sigma, \tau \mid \tau^\sigma = \tau^2 \rangle$, conjugation preserves the order of τ . Hence τ and τ^2 have the same order, so τ has odd order. It follows that $\tau^{\omega_2} = 1$.

After killing x_0, x_1 , the wild relation is exactly $\tau^{\omega_2} = 1$, so it is redundant. This proves the tame quotient claim. The displayed assignment to $\mathbf{Z}_2$ respects both relators: every factor of r_R has unramified image zero. It is surjective because σ maps to 1. $\square$

The group $\mathcal{T}$ has no nontrivial normal pro-2 subgroup by the argument of [1, Lemma 3.3]. Since W_R is pro-2, it follows that

$$W_R = O_2(\Gamma_R).$$

Thus the tame half of the boundary in [1, §4] is unchanged.

3 The maximal pro-2 quotient

3.1 The relator and its cup form

Lemma 3.1. *The maximal pro-2 quotient of Γ_R is*

$$D_R = \langle s, x, y \mid r_2 = (x^s)^{-1} x^{-3} y^2 [y, y^s] = 1 \rangle_{\text{pro-2}}, \quad (3.1)$$

where $s = \sigma$, $x = x_0$, and $y = x_1$.

Proof. In a pro-2 quotient, the closure of $\langle \tau \rangle$ is procyclic. Conjugation by σ acts on it as squaring. Squaring is not an automorphism of a nontrivial procyclic pro-2 group, so $\tau = 1$. Profinite exponentiation by ω_2 is the identity on pro-2 elements. Substitution in r_R gives equation (3.1). Conversely, every finite quotient of equation (3.1) is R -admissible, so this is the maximal pro-2 quotient. $\square$

Let $\mathcal{D}_n$ denote the n -th Zassenhaus subgroup of the free pro-2 group on s, x, y .

Lemma 3.2. *The degree-two initial form of r_2 is*

$$y^2 + [x, s] \in \mathcal{D}_2 / \mathcal{D}_3.$$

Consequently D_R is a rank-three Demushkin group, and its cup–Bockstein matrix in the basis dual to s, x, y is

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (3.2)$$

Proof. Since $x^s = x[x, s]$, one has

$$r_2 = [x, s]^{-1} x^{-4} y^2 [y, y^s].$$

Now $x^{-4} \in \mathcal{D}_4$. Also $y^s = y[y, s]$ with $[y, s] \in \mathcal{D}_2$, whence $[y, y^s] \in \mathcal{D}_3$. This gives the asserted initial form.

The presentation is minimal and has one relation, so $\dim_{\mathbf{F}_2} H^2(D_R, \mathbf{F}_2) = 1$. The standard relation–cup formula gives equation (3.2), which is nonsingular. This is exactly the defining cohomological condition for a Demushkin group; compare [2, §1]. $\square$

Abelianizing the relation and putting $t = \bar{y} - 2\bar{x}$ gives

$$B_R := D_R^{\text{ab}} = \langle \bar{s}, \bar{x}, \bar{y} \mid -4\bar{x} + 2\bar{y} = 0 \rangle_{\mathbf{Z}_2}, \quad (3.3)$$

$$t = \bar{y} - 2\bar{x}, \quad (3.4)$$

$$B_R = C_2 t \oplus \mathbf{Z}_2 \bar{s} \oplus \mathbf{Z}_2 \bar{x}, \quad \nu_R(t, \bar{s}, \bar{x}) = (0, 1, 0). \quad (3.5)$$

3.2 The canonical orientation

Agreement as an unmarked Demushkin group is insufficient: the tame and pro-2 quotients must have the same map to the full quotient $\mathbf{Z}_2$. We therefore compute the canonical Demushkin orientation integrally.

Proposition 3.3. *Let $\chi_R : D_R \rightarrow \mathbf{Z}_2^\times$ be the canonical Demushkin orientation and write*

$$S = \chi_R(s), \quad X = \chi_R(x), \quad Y = \chi_R(y).$$

Then

$$Y = -X^2, \quad X^3 + 2X^2 + 1 = 0, \quad S = -\frac{X^3}{X^2 + X + 1}. \quad (3.6)$$

The polynomial $Z^3 + 2Z^2 + 1$ has a unique root in $\mathbf{Z}_2^\times$, and

$$X \equiv 5 \pmod{16}, \quad S \equiv 13 \pmod{16}.$$

In particular,

$$\text{im } \chi_R = \{\pm 1\} \times (1 + 4\mathbf{Z}_2).$$

Proof. We apply Labute's characterization of the canonical orientation [2, Theorem 4]. For a character χ of the free pro-2 group, let D be a crossed derivation into the rank-one module defined by χ :

$$D(gh) = Dg + \chi(g)Dh, \quad D(g^{-1}) = -\chi(g)^{-1}Dg.$$

Such derivations descend through the relation precisely when $D(r_2) = 0$ for arbitrary values of Ds, Dx, Dy .

First,

$$D(x^s) = S^{-1}Dx + S^{-1}(X - 1)Ds.$$

For a commutator of elements having character values G, H ,

$$D[g, h] = G^{-1}(H^{-1} - 1)Dg + H^{-1}(1 - G^{-1})Dh. \quad (3.7)$$

Using equation (3.7) with $g = y, h = y^s$, and collecting the three free coefficients in

$$r_2 = (x^s)^{-1}x^{-3}y^2[y, y^s]$$

gives

$$Y^2 = X^4, \quad (3.8)$$

$$0 = -(X^{-1}S^{-1} + X^{-2} + X^{-3} + X^{-4}), \quad (3.9)$$

$$0 = S^{-1} \left[-X^{-1}(X - 1) + \frac{(Y - 1)^2}{Y^2} \right], \quad (3.10)$$

$$0 = X^{-4}(1 + Y) + Y^{-1}(1 - Y^{-1})(S^{-1} - 1). \quad (3.11)$$

For completeness, these are respectively the character relation and the coefficients of Dx, Ds, Dy .

By equation (3.8), $Y = \pm X^2$, since the only square roots of 1 in $\mathbf{Z}_2^\times$ are ± 1 . Equation (3.9) gives

$$S = -\frac{X^3}{X^2 + X + 1}; \quad (3.12)$$

the denominator is a unit.

Suppose first that $Y = X^2$. After clearing denominators, [equation \(3.10\)](#) becomes

$$(X - 1)(X^2 - X - 1) = 0.$$

The second factor is odd and therefore cannot vanish in $\mathbf{Z}_2$, so $X = 1$. Equation [\(3.11\)](#) then has left side 2, a contradiction. Thus this branch is impossible.

For $Y = -X^2$, equation [\(3.10\)](#) becomes

$$X^3 + 2X^2 + 1 = 0.$$

After [equation \(3.12\)](#) is substituted, the left side of [equation \(3.11\)](#) is

$$\frac{(X + 1)(X^3 + 2X^2 + 1)}{X^7},$$

so [equation \(3.11\)](#) is automatic. This proves [equation \(3.6\)](#).

Every 2-adic unit is congruent to 1 modulo 2. For $f(Z) = Z^3 + 2Z^2 + 1$, one has

$$f(1) \equiv 0 \pmod{2}, \quad f'(1) = 7 \equiv 1 \pmod{2}.$$

Hensel's lemma therefore gives a unique unit root. Lifting to modulus 16 gives $X \equiv 5$, and [equation \(3.12\)](#) gives $S \equiv 13$. Hence

$$v_2(X - 1) = v_2(S - 1) = 2.$$

The 2-adic logarithm identifies $1 + 4\mathbf{Z}_2$ with $4\mathbf{Z}_2$; an element whose difference from 1 has valuation 2 is a topological generator. Finally, from [equation \(3.4\)](#),

$$\chi_R(t) = YX^{-2} = -1.$$

Thus the image of χ_R is exactly $\{\pm 1\} \times (1 + 4\mathbf{Z}_2)$. □

Corollary 3.4. *The group D_R is abstractly isomorphic to*

$$D_0 = \langle A, S_0, Y_0 \mid A^2 S_0^4 [S_0, Y_0] = 1 \rangle_{\text{pro-2}}, \quad (3.13)$$

and hence to $G_{\mathbf{Q}_2}(2)$.

Proof. The abelianization [equation \(3.3\)](#) has torsion subgroup of order 2, so the Labute invariant is $q = 2$. The group has odd rank 3, and [proposition 3.3](#) shows that the secondary orientation depth is $f = 2$. Labute's odd-rank classification [2, Theorems 4 and 8] therefore gives the normal form [equation \(3.13\)](#). The same normal form is the maximal pro-2 Galois group of $\mathbf{Q}_2$; see [1, Lemma 3.4]. □

3.3 Matching the full unramified character

Let

$$t_0 = \bar{A} + 2\bar{S}_0, \quad \eta = (-3)^{-1}.$$

The standard abelianization and its two characters are

$$D_0^{\text{ab}} = C_2 t_0 \oplus \mathbf{Z}_2 \bar{S}_0 \oplus \mathbf{Z}_2 \bar{Y}_0, \quad (3.14)$$

$$\nu_0(t_0, \bar{S}_0, \bar{Y}_0) = (0, 1, 0), \quad (3.15)$$

$$\chi_0(t_0, \bar{S}_0, \bar{Y}_0) = (-1, 1, \eta). \quad (3.16)$$

These are [1, (3.4)–(3.6)].

Proposition 3.5. *There is an isomorphism of fully marked pro-2 groups*

$$(D_R, \nu_R) \cong (G_{\mathbf{Q}_2}(2), \nu_{\text{ur}}).$$

Proof. Since X, S both topologically generate $1 + 4\mathbf{Z}_2$, there is a unique $b \in \mathbf{Z}_2^\times$ such that $S = X^b$. In the additive notation of B_R , put

$$k = \bar{s} - b\bar{x}.$$

Equations (3.5) and (3.6) give

$$B_R = C_2 t \oplus \mathbf{Z}_2 k \oplus \mathbf{Z}_2 \bar{x}, \quad \nu_R(t, k, \bar{x}) = (0, 1, 0), \quad \chi_R(t, k, \bar{x}) = (-1, 1, X).$$

The element $\eta = (-3)^{-1}$ also topologically generates $1 + 4\mathbf{Z}_2$, so there is a unique unit $u \in \mathbf{Z}_2^\times$ with $\eta^u = X$. Therefore

$$\phi_{\text{ab}} : B_R \xrightarrow{\sim} D_0^{\text{ab}}, \quad t \mapsto t_0, \quad k \mapsto \bar{S}_0, \quad \bar{x} \mapsto u\bar{Y}_0 \quad (3.17)$$

preserves both ν and the canonical orientation.

Choose an abstract isomorphism $f : D_R \rightarrow D_0$ from [corollary 3.4](#). Canonical orientations are functorial, so both f_{ab} and ϕ_{ab} preserve orientation. Hence $\phi_{\text{ab}} f_{\text{ab}}^{-1}$ is an orientation-preserving automorphism of D_0^{ab} . By [\[1, Proposition 3.9\]](#), every such automorphism lifts to an automorphism γ of D_0 . Replacing f by γf gives an isomorphism whose abelianization is ϕ_{ab} , and therefore preserves the full $\mathbf{Z}_2$ -valued unramified character. Finally use the marked local normalization [\[1, Proposition 1.1\]](#). $\square$

Combining [lemma 2.1](#) and [proposition 3.5](#), the tame and pro-2 boundary

$$\mathcal{T} \times_{\mathbf{Z}_2} G_{\mathbf{Q}_2}(2)$$

is the same for Γ_R and $G_{\mathbf{Q}_2}$. This verifies the structural input to [\[1, §4\]](#).

4 The evaluated Fox complex

Fix a finite boundary-compatible lower quotient C , a lower map $\rho : \Gamma_R \twoheadrightarrow C$, and an elementary $\mathbf{F}_2[C]$ -module V . On a simple composition factor, the wild normal subgroup acts trivially by [\[1, Lemma 5.12\]](#); hence the action is tame. Write

$$\mathbf{S} = \rho(\sigma)|_V, \quad \mathbf{T} = \rho(\tau)|_V, \quad \mathbf{U} = \mathbf{S}^{\omega_2}|_V.$$

If e is the odd order of $\mathbf{T}$, put

$$P = 1 + \mathbf{T} + \cdots + \mathbf{T}^{e-1}.$$

This is the projector onto $V^{\mathbf{T}}$. Write variations of (σ, τ, x_0, x_1) as $(a, b, c, d) \in V^4$.

Proposition 4.1 (Evaluated Jacobian). *On a simple tame module the differential obtained by differentiating the two relators is*

$$d_{R, \rho, V}^1 = \begin{pmatrix} \mathbf{S}^{-1}(1 + \mathbf{T}) & \mathbf{S}^{-1} + 1 + \mathbf{T} & 0 & 0 \\ 0 & P & P + \mathbf{S}^{-1} & 0 \end{pmatrix}. \quad (4.1)$$

Thus it is exactly the matrix in [\[1, \(5.4\)\]](#) after interchanging the two wild columns.

Proof. The tame row is unchanged from [1, (5.5)]:

$$L_t = S^{-1}(1 + T)a + (S^{-1} + 1 + T)b.$$

For the wild row, the action of x_0, x_1 on V is trivial. Consequently

$$D((x_0^\sigma)^{-1}) = S^{-1}c.$$

For $g = x_0^{-3}\tau$, one has

$$Dg = (-3)c + b = c + b$$

in characteristic 2, and g acts on V as T . The profinite power/norm rule used in [1, Lemma 5.5] gives

$$D(g^{\omega_2}) = P(c + b).$$

Finally,

$$D(x_1^2) = (1 + 1)d = 0,$$

and both entries of $[x_1, x_1^{\sigma_2}]$ act trivially, so its first derivative is zero. Therefore

$$L_w = Pb + (P + S^{-1})c,$$

which proves [equation \(4.1\)](#). □

Lemma 4.2 (Simple normal forms). *Let V be nontrivial and simple. Every class in the degree-one cohomology of the word complex has a unique representative*

$$(a, b, c, d) = (0, 0, 0, d).$$

The Jacobian is surjective, and

$$H_R^0(V) = H_R^2(V) = 0, \quad \dim_{\mathbf{F}_2} H_R^1(V) = \dim_{\mathbf{F}_2} V.$$

Proof. The normal subgroup generated by tame inertia makes V^T a submodule. Simplicity gives two cases.

If $T = 1$, then $P = 1$. The tame and wild rows of [equation \(4.1\)](#) first give

$$b = 0, \quad (1 + S^{-1})c = 0.$$

Since V is nontrivial simple, $S - 1$ is invertible; hence $c = 0$, and the unique coboundary with $(S - 1)v = a$ kills a .

If $V^T = 0$, then $P = 0$ and $T - 1 = 1 + T$ is invertible. The wild row gives $c = 0$. Subtracting the coboundary of $(T - 1)^{-1}b$ kills b , after which the tame row forces $a = 0$. Uniqueness follows in both cases from the same invertibility statements. Surjectivity of the two relation rows and the cohomological dimension assertions follow immediately. □

Lemma 4.3 (Trivial coefficient). *For $V = \mathbf{F}_2$ with trivial action,*

$$d_R^1(a, b, c, d) = (b, b).$$

Thus $H_R^1(\mathbf{F}_2)$ has coordinates (a, c, d) , and its scalar cup–Bockstein form is

$$\langle (a, c, d), (a', c', d') \rangle = ac' + ca' + dd'. \quad (4.2)$$

Proof. In [equation \(4.1\)](#), set $S = T = P = 1$. This gives the displayed differential. Every homomorphism to $\mathbf{F}_2$ factors through the maximal pro-2 quotient. The initial form of [lemma 3.2](#) therefore gives [equation \(4.2\)](#). Equivalently, it is the form with matrix [equation \(3.2\)](#). □

5 Fox–Heisenberg duality

5.1 The Stokes endpoint

The chain map in [1, §5.2] is obtained by tracing the two relation-defect coordinates to one scalar. For a new word, the only endpoint issue is the finite-word Stokes condition.

Lemma 5.1. *Modulo 2, the exponent vectors of r_t and r_R , in the ordered basis (σ, τ, x_0, x_1) , are both*

$$(0, 1, 0, 0).$$

Their sum is zero. Hence the trace

$$(u_t, u_w) \mapsto u_t + u_w$$

satisfies the endpoint condition of the finite-word Stokes formula [1, Lemma 5.7 and Proposition 5.8].

Proof. For $r_t = (\tau^\sigma)^{-1} \tau^2$, the τ -exponent is $-1 + 2 \equiv 1 \pmod{2}$.

Choose any finite integer exponent representing ω_2 at the finite level under consideration. It is odd. In r_R , the first factor contributes one x_0 modulo 2, while $(x_0^{-3} \tau)^{\omega_2}$ contributes one x_0 and one τ . The two x_0 -contributions cancel. The square x_1^2 and the commutator have zero exponent vector modulo 2. Thus only the τ -coordinate remains. $\square$

5.2 The mixed Hessian

Proposition 5.2. *Let V be nontrivial simple and represent a degree-one class by $(0, 0, 0, d)$ as in lemma 4.2. Represent the dual class by $\lambda \in V^\vee$ in the corresponding x_1 -supported normal form. The degree-one pairing induced by the Fox–Heisenberg chain map is*

$$(d, \lambda) \mapsto \begin{cases} \lambda(d), & T = 1, \\ \lambda((1 + U + U^{-1})d), & V^T = 0. \end{cases} \quad (5.1)$$

Both operators are invertible.

Proof. On the normalized representatives, the factors $(x_0^\sigma)^{-1}$ and $(x_0^{-3} \tau)^{\omega_2}$ have no varying coordinate. The square x_1^2 contributes the usual Heisenberg diagonal $\lambda(d)$. The class-two commutator identity applied to $[x_1, x_1^{\sigma^2}]$ contributes

$$\lambda(Ud) + \lambda(U^{-1}d).$$

If $T = 1$, then $U = 1$ on a nontrivial simple module, and these two commutator terms cancel. If $V^T = 0$, they give the second line of equation (5.1).

In the ramified case, U has 2-power order, so its minimal polynomial divides a power of $Z + 1$. Since $Z^2 + Z + 1$ is coprime to $Z + 1$,

$$1 + U + U^{-1} = U^{-1}(U^2 + U + 1)$$

is invertible. The unramified operator is the identity. $\square$

Proposition 5.3 (Candidate deformation duality). *For every elementary, possibly nonsplit, $\mathbf{F}_2[C]$ -module V , the finite-word complex of Γ_R is naturally self-dual in the sense of [1, Proposition 5.15]. In particular, it has the same obstruction dimensions, lift multiplicities, perfect pairings, and adjoint connecting homomorphisms as the local cochain complex of $G_{\mathbf{Q}_2}$.*

Proof. By lemma 5.1, the finite-word Stokes construction gives a natural chain map. For the trivial simple module it is a quasi-isomorphism by the nonsingularity of equation (4.2). For every nontrivial simple module, lemma 4.2 and proposition 5.2 show that the degree-one pairing is perfect and that $H^0 = H^2 = 0$. Hence the cone of the chain map is acyclic on all simple modules.

The word complex, its dual, and the chain map are exact coefficient functors. Applying the degreewise short exact cone sequence along a composition series and using two-out-of-three extends acyclicity to every elementary module. This is exactly the cone devissage argument of [1, Lemma 5.11 and Proposition 5.15]. Naturality of the chain map gives adjointness of connecting morphisms. Comparison with the local complex then uses local Tate duality as in [1, Proposition 5.16 and Corollary 5.17]. $\square$

6 Quadratic determinant obstructions

Let V be a nontrivial self-dual simple module, and let

$$q : V \longrightarrow \mathbf{F}_2$$

be a nonzero invariant nonsingular quadratic form with polar form b_q . Let κ_q^0 be the normalized base determinant class of [1, Lemma 6.3].

Proposition 6.1 (Base word expansion). *On the normalized class $(0, 0, 0, d)$, evaluation of Roe's wild word gives*

$$Q_R^0(d) = \begin{cases} q(d), & T = 1, \\ q(d) + b_q(d, U^{-1}d), & V^T = 0. \end{cases} \quad (6.1)$$

In the ramified case its polar form is

$$b_R(d, d') = b_q(d, (1 + U + U^{-1})d'). \quad (6.2)$$

Proof. As in the mixed calculation, only the final two factors vary on the normalized representative. The square x_1^2 contributes $q(d)$. The complete class-two commutator calculation for $[x_1, x_1^{\sigma_2}]$ contributes

$$b_q(d, U^{-1}d).$$

No equivariance of a chosen section is being assumed here: changing the section alters the factor set by a normalized coboundary, and the complete lifted multiplication formula of [1, (6.8)] yields the same cohomology class.

If $T = 1$, then $U = 1$ and $b_q(d, d) = 0$, giving the first line. Polarizing the second line and using U -invariance and symmetry of b_q gives equation (6.2). $\square$

The formula in equation (6.1) is exactly [1, (6.25)], with the normalized wild coordinate renamed from c to d . Consequently all of the candidate-side Arf computations in [1, §6.2.1] apply without change.

Corollary 6.2 (Gauss signs). *If $n = \dim_{\mathbf{F}_2} V$, then*

$$\#(Q_R^0)^{-1}(0) = \begin{cases} 2^{n-1} - 2^{n/2-1}, & V \text{ unramified,} \\ 2^{n-1} + 2^{n/2-1}, & V \text{ ramified.} \end{cases}$$

These are the same zero counts, and hence the same Gauss signs, as the local determinant obstruction for $G_{\mathbf{Q}_2}$.

Proof. For unramified V , [equation \(6.1\)](#) is the invariant quadratic form q ; the Hermitian-line calculation of [1, Proposition 6.9] gives the negative sign. In the ramified case, [equation \(6.1\)](#) is the Wall-twisted form used there, so [1, Lemmas 6.6 and 6.8] give Arf invariant zero and the positive sign. Equality with the local signs is [1, Proposition 6.18]. $\square$

For a general determinant class

$$\kappa = \kappa_q^0 + \Gamma_{\gamma_\kappa} + \inf \delta_\kappa,$$

the difference from the base form is the natural linear pairing term plus the inflated scalar term, exactly as in [1, (6.7) and (6.27)]. This assertion is formal in the determinant class and the natural chain pairing; no additional word calculation is required. Thus the affine transgression and shear corrections of [1, §6.5] also carry over.

7 Passage to every finite target

Proposition 7.1. *For every finite lower quotient, every boundary-compatible exact-image map, and every elementary characteristic-2 coefficient module, the sources Γ_R and $G_{\mathbf{Q}_2}$ have:*

1. *the same obstruction-space dimensions and the same number of lifts when the obstruction vanishes;*
2. *natural perfect pairings for all short exact coefficient sequences, with adjoint connecting maps;*
3. *the same scalar cup–Bockstein form;*
4. *for every invariant nonsingular quadratic form on a self-dual simple head, the same base determinant Gauss sign.*

Proof. Items (1) and (2) are [proposition 5.3](#); item (3) is [lemma 4.3](#); and item (4) is [corollary 6.2](#). The marked maximal pro-2 input is [proposition 3.5](#), and the common tame–pro-2 boundary follows from [lemma 2.1](#). This is the analogue for Γ_R of the complete source interface [1, Corollary 6.19]. $\square$

Proof of theorem 1.2. The target-side operations in [1, §§7–9] are restrictions, pushouts, exact-image partitions, finite Fourier transforms, and central cover constructions internal to the finite target. By [1, Theorem 8.17], the exact-image recursion depends on the profinite source only through the common boundary and the source interface listed in [proposition 7.1](#). Replacing the original candidate source by Γ_R therefore leaves the recursion unchanged.

The boundary-framed induction of [1, Theorem 4.2 and §9] gives equal exact-image counts for Γ_R and $G_{\mathbf{Q}_2}$. Removing the framing as in [1, §10] yields

$$|\mathrm{Sur}(\Gamma_R, H)| = |\mathrm{Sur}(G_{\mathbf{Q}_2}, H)|$$

for every finite group H . The group Γ_R is topologically finitely generated. The one-sided profinite reconstruction lemma [1, Lemma 2.5] now gives $\Gamma_R \cong G_{\mathbf{Q}_2}$. $\square$

8 Audit summary and remaining formalization work

The proof above checks all features of Roe’s relator that are visible to the source interface:

- its tame specialization is redundant;
- its maximal pro-2 quotient has the correct Demushkin form, orientation image, and full $\mathbf{Z}_2$ -marking;
- its evaluated Fox row is the paper’s row with the two wild columns interchanged;
- its mod-2 exponent vector satisfies the Stokes endpoint;
- its mixed Heisenberg Hessian and determinant quadratic form are exactly those used in the paper after the same wild-coordinate interchange.

This is not an explicit Nielsen or Tietze transformation between Roe’s presentation and the longer presentation in [1, Theorem 1.2]. No such transformation is needed for the finite-quotient proof. The result instead shows that the two sources have the same exact-image counts against every finite target, from which abstract profinite isomorphism follows.

For a machine-checked version, the genuinely new formal obligations are short and well isolated:

1. formalize [equations \(3.8\) to \(3.11\)](#) and the Hensel root calculation;
2. formalize the wild Fox row in [equation \(4.1\)](#);
3. formalize the $x_1^2[x_1, x_1^{\sigma_2}]$ mixed and quadratic Heisenberg expansions;
4. instantiate the generic finite-word Stokes theorem using [lemma 5.1](#).

The local calculations and finite-target induction need no new candidate-specific mathematics.

References

- [1] D. Roe and D. Turturean, *A Presentation of the Absolute Galois Group of $\mathbf{Q}_2$* , 2026. <https://roed314.github.io/gq2/paper/paper.html>.
- [2] J. P. Labute, *Classification of Demushkin Groups*, Canadian Journal of Mathematics **19** (1967), 106–132. <https://doi.org/10.4153/CJM-1967-007-8>.
- [3] D. Roe, *Presentations of $G_{\mathbf{Q}_2}$: David Roe’s candidate*, 2026. <https://roed314.github.io/gq2/presentations/#candidate-roe>.